US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250281031
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	HASE; Hidenosuke

ENDOSCOPE

Abstract

An endoscope includes a bending tube extending along a longitudinal axis from a proximal end side to a distal end side, a wire that bends the bending tube, a first sheath through which the wire is passed, and a plate body having an annular shape and disposed in the proximal end side of the bending tube. The plate body includes: a first annular surface facing distally; a second annular surface facing proximally and intersecting the longitudinal axis; and a hole penetrating the first annular surface and the second annular surface and through which the wire is passed, and at least a portion of the first sheath is disposed proximally relative to the second annular surface and a distal end of at least the portion of the first sheath is configured to push a periphery of the hole on the second annular surface.

Inventors: **HASE; Hidenosuke (Tokyo, JP)**

Applicant: **OLYMPUS MEDICAL SYSTEMS CORP.** (Tokyo, JP)

Family ID: **96810573**

Assignee: **OLYMPUS MEDICAL SYSTEMS CORP. (Tokyo, JP)**

Appl. No.: **19/070868**

Filed: **March 05, 2025**

Related U.S. Application Data

us-provisional-application US 63561925 20240306

Publication Classification

Int. Cl.: **A61B1/005** (20060101)

U.S. Cl.:

CPC **A61B1/0057** (20130101);

Background/Summary

RELATED APPLICATION DATA [0001] This application is based on and claims priority under 35 U.S.C. § 119 to U.S. Provisional Application No. 63/561,925, filed on Mar. 6, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND
1. Field
[0002] The present disclosure relates to an endoscope including a bending tube that is bent by pulling or loosening a wire.
2. Related Art
[0003] Endoscopes including a bending tube (also referred to as a bending portion) have been proposed in the past. The bending tube changes the direction of a distal end portion by a bending wire being pulled or loosened, thereby changing the direction of observation with the endoscope. The bending wire is passed through from an operation portion of the endoscope to the bending tube.
[0004] In a flexible tube, the bending wire is passed through a sheath that guides the wire. In this case, a configuration may be adopted to reduce the friction induced between the sheath and the wire.
[0005] For example, if watertightness in the insertion portion can be ensured, lubricants such as silicone oil and carbon powder may be used to reduce the friction.
[0006] If watertightness in the insertion portion cannot be ensured, an inner sheath may be provided that has less friction with the wire than the sheath. At this time, the wire is passed through the sheath while being in a state of being passed through the inner sheath.
[0007] For example, Japanese Patent No. 6028136 describes using, as a sheath through which a bending wire is to be passed, a coil sheath formed by winding a core wire in a coil shape. Specifically, the bending wire is passed through an inner coil sheath, and is further passed through an outer coil sheath. A distal end portion of the outer coil sheath is fixed to an inner surface of the flexible tube on a distal end side, and a proximal end surface of the outer coil sheath is fixed to a fixing plate in an operation portion.
SUMMARY
[0008] An endoscope according to an aspect of the present disclosure includes a bending tube extending along a longitudinal axis from a proximal end side to a distal end side, a wire configured to bend the bending tube, a first sheath through which the wire is passed, and a plate body having an annular shape and disposed in the proximal end side of the bending tube. The plate body includes: a first annular surface facing distally; a second annular surface facing proximally and intersecting the longitudinal axis; and a hole penetrating the first annular surface and the second annular surface and through which the wire is passed, and at least a portion of the first sheath is disposed proximally relative to the second annular surface, and a distal end of at least the portion of the first sheath is configured to push a periphery of the hole on the second annular surface.
[0009] An endoscope according to an aspect of the present disclosure includes a bending tube extending along a longitudinal axis from a proximal end side to a distal end side, a wire configured to bend the bending tube, a first sheath through which the wire is passed, and a plate body having an annular shape and disposed in the proximal end side of the bending tube. The plate body includes: a first annular surface facing distally; a second annular surface facing proximally and intersecting the longitudinal axis; and a hole penetrating the first annular surface and the second annular surface and through which the wire is passed, and at least a portion of the first sheath is

disposed proximally relative to the second annular surface, and a distal end of at least the portion of the first sheath is configured to produce a frictional force in a direction intersecting the longitudinal axis between the distal end of at least the portion and a periphery of the hole on the second annular surface.

[0010] An endoscope according to an aspect of the present disclosure includes a bending tube extending along a longitudinal axis from a proximal end side to a distal end side, a wire configured to bend the bending tube, a first sheath, a second sheath passed through the first sheath and through which the wire is passed, and a plate body having an annular shape and disposed in the proximal end side of the bending tube. The plate body includes: a first annular surface facing distally; a second annular surface facing proximally; and a hole penetrating the first annular surface and the second annular surface and through which the wire is passed, at least a portion of the first sheath is disposed proximally relative to the second annular surface, and a distal end of at least the portion of the first sheath is configured to push a periphery of the hole on the second annular surface, and the wire and the second sheath are passed through the hole and a distal end of the second sheath is disposed distally relative to the hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a perspective view showing a configuration of an endoscope of a first embodiment of the present disclosure with some portions omitted.

[0012] FIG. **2** is a cross-sectional view showing a configuration of a bending wire at a connection part between a bending tube and a flexible tube in the first embodiment.

[0013] FIG. **3** is a partial perspective view showing a configuration of the bending wire, an inner sheath, a sheath, and a plate body in the first embodiment.

[0014] FIG. **4** is a cross-sectional view of the bending tube near the plate body, taken orthogonal to a longitudinal axis, in the first embodiment.

[0015] FIG. **5** is a cross-sectional view of a modification of the plate body in the first embodiment, when cut in a plane parallel to the longitudinal axis.

[0016] FIG. **6** is a cross-sectional view showing, from an oblique direction, a configuration of a second plate body disposed on a distal end side of the bending tube in the first embodiment.

[0017] FIG. **7** is a cross-sectional view showing a wire that is passed through four holes in the second plate body in the first embodiment.

[0018] FIG. **8** is a cross-sectional view showing two wires that are respectively passed through two holes in the second plate body in the first embodiment.

[0019] FIG. **9** is a perspective view showing a fixing body disposed in an operation portion in the first embodiment.

[0020] FIG. **10** is a cross-sectional view showing a portion of a structure of the fixing body in the first embodiment.

[0021] FIG. **11** is a view showing how a position to cut the sheath is set according to an actual length of a flexible tube body in the first embodiment.

[0022] FIG. **12** is a side view showing an operation mechanism disposed in the operation portion in the first embodiment.

[0023] FIG. **13** is a partial perspective view showing the operation mechanism of the first embodiment from a direction of an arrow A**1** in FIG. **12**.

[0024] FIG. **14** is a view for illustrating a process of fixing the bending wire to the operation mechanism in the first embodiment.

[0025] FIG. **15** is a view showing that the sheath is held in a state of pressing the plate body and the fixing body in the first embodiment.

[0026] FIG. **16** is a cross-sectional view showing a relationship between the sheath and the wire when the inner sheath is not provided in the first embodiment.
[0027] FIG. **17** is a view showing a configuration example of the sheath in a second embodiment of the present disclosure.
[0028] FIG. **18** is a view showing a first example of a swing angle of a wire in a third embodiment of the present disclosure.
[0029] FIG. **19** is a view showing a second example of the swing angle of the wire in the third embodiment.
[0030] FIG. **20** is a view showing a configuration example of a sheath in a fourth embodiment of the present disclosure.
[0031] FIG. **21** is diagram showing a configuration example of a part of a second annular surface of a plate body, where a distal end of a sheath abuts, in a fifth embodiment of the present disclosure.

DETAILED DESCRIPTION

[0032] In general, a sheath through which a bending wire is passed is subjected to a pulling force load when the bending wire is pulled to bend a bending tube. For this reason, the sheath is firmly fixed to a distal end side of a flexible tube or a proximal end side of the bending tube by welding or the like, for example.
[0033] However, welding or the like is not suitable for manufacturing inexpensive endoscopes because skilled workers must use jigs to perform the work.
[0034] According to embodiments described below, an endoscope can be provided including a sheath through which a bending wire is passed, suitable for manufacturing inexpensive endoscopes.
[0035] The following describes embodiments of the present disclosure with reference to the drawings. However, the present disclosure is not limited by the embodiments described below.
[0036] Note that, in the description of the drawings, the same or corresponding elements are attached with the same symbol as appropriate. It should be noted that the drawings are schematic, and the relationship of lengths of each element, the ratio of lengths of each element, the quantity of each element, etc. within a single drawing may differ from reality for the sake of brevity of explanation. Furthermore, even among a plurality of drawings, they may contain portions that differ from each other in terms of length relationships, ratios, quantities, etc.

FIRST EMBODIMENT

[0037] FIGS. **1** to **16** show a first embodiment of the present disclosure. FIG. **1** is a perspective view showing a configuration of an endoscope **1** of a first embodiment with some portions omitted.
[0038] The endoscope **1** is an insertion instrument including a section to be inserted into a subject. The subject can be a living body such as a human or an animal, or a non-living body such as a machine or a building.
[0039] The endoscope **1** includes an insertion portion **2**, an operation portion **3**, and a universal cable **4**.
[0040] The insertion portion **2** is a section configured to be inserted into the subject. The insertion portion **2** includes a distal end portion **2***a*, a bending tube **2***b*, and a flexible tube **2***c*, in order from the distal end side to the proximal end side.
[0041] In the distal end portion **2***a*, an image pickup device, an image pickup optical system, an illumination optical system, a nozzle **22***a* (see FIG. **8**), a channel distal-end-side opening **21***a* (see FIG. **8**), etc., are disposed. The illumination optical system irradiates the subject with illumination light. The image pickup optical system forms an optical image of the subject. The image pickup device photoelectrically converts the optical image formed by the image pickup optical system, to generate an image pickup signal. The nozzle **22***a* discharges gas and liquid fed via a gas/liquid feeding channel **22** (see FIGS. **4**, **6**, **8**, etc.) to an observation window at a distal end of the image pickup optical system. The channel distal-end-side opening **21***a* is an opening on the distal end side of a treatment instrument channel **21** (see FIGS. **2**, **4**, **6**, etc.).
[0042] The bending tube **2***b* is a section that is bendable, for example, in four directions (up, down,

left, and right). The bending tube **2***b* is also referred to as a bending portion. The bending tube **2***b* is provided along a longitudinal axis O (see FIGS. **2**, **3**, etc.) that extends from the proximal end side to the distal end side of the endoscope **1**.

[0043] The flexible tube **2***c* is a tube portion having flexibility, is also referred to as a flexible tube portion. The flexible tube **2***c* is provided along the longitudinal axis O, on the proximal end side of the bending tube **2***b*. Note that here, an example is given in which the endoscope **1** is a flexible endoscope including the flexible tube **2***c*. However, the endoscope **1** can also be a rigid endoscope in a form where the part corresponding to the flexible tube **2***c* is rigid.

[0044] The operation portion **3** is provided on the proximal end side of the flexible tube **2***c* of the insertion portion **2**. The operation portion **3** is a section for the user to operate the endoscope **1**. The operation portion **3** includes a grasping portion **5**, a bending operation knob **6**, various operation switches **7**, a gas/liquid feeding button **8***a*, a suction button **8***b*, and a treatment-instrument insertion opening **9**.

[0045] The grasping portion **5** is a section where the user grasps the endoscope **1** with the palm.

[0046] The bending operation knob **6** is an operation device for operating the bending of the bending tube **2***b*. The bending operation knob **6** is operated, for example, using the thumb of the hand grasping the grasping portion **5**. When the bending operation knob **6** is operated, the bending wire **11** (see FIGS. **2**, **3**, etc.) is pulled and the bending tube **2***b* is bent.

[0047] The bending operation knob **6** includes an up-down bending operation knob **6***a* and a left-right bending operation knob **6***b*. The up-down bending operation knob **6***a* is operated to bend the bending tube **2***b* in the up and down directions. The left-right bending operation knob **6***b* is operated to bend the bending tube **2***b* in the left and right directions.

[0048] When the bending tube **2***b* is bent, the direction of the distal end portion **2***a* changes. This changes the image pickup direction by the image pickup device and the image pickup optical system and the illumination direction of the illumination light by the illumination optical system. The bending tube **2***b* is bent also to improve insertability of the insertion portion **2** in the subject.

[0049] The operation switches **7** include a button switch relating to image pickup. Specific examples of the operation switches **7** are button switches such as a freeze button to pause the monitor screen and a release button to pick up a still image.

[0050] The gas/liquid feeding button **8***a* is a button for an operation of feeding gas and liquid to the observation window of the distal end portion **2***a*. The liquid feeding cleans the observation window, and the gas feeding blows away the liquid after the cleaning. The gas feeding and liquid feeding are performed via the gas/liquid feeding channel **22**.

[0051] The suction button **8***b* is a button for performing an operation of suctioning the inside of the subject from the distal end portion **2***a*. The suction from inside the subject is performed, for example, via the treatment instrument channel **21**, which also serves as a suction channel. When the suction operation is performed, for example, liquid or mucous membrane is suctioned from inside the subject.

[0052] The treatment-instrument insertion opening **9** is an opening on the proximal end side of the treatment instrument channel **21**. A treatment instrument, such as a forceps, is inserted from the treatment-instrument insertion opening **9** into the treatment instrument channel **21**. A distal end portion of the treatment instrument is guided from the treatment instrument channel **21** to the channel distal-end-side opening **21***a*, and protrudes into the subject. The distal end portion of the protruded treatment instrument is used to perform various treatments on the subject.

[0053] The universal cable **4** is extended from, for example, a side surface on the proximal end side of the operation portion **3**. A connector is provided at an extended end of the universal cable **4**. The connector connects the endoscope **1** to an endoscope processor (video processor), a light source apparatus, a suction pump, a liquid feeding tank, etc.

[0054] FIG. **2** is a cross-sectional view showing a configuration of the bending wire **11** at a connection part between the bending tube **2***b* and the flexible tube **2***c* in the first embodiment. FIG.

3 is a partial perspective view showing a configuration of the bending wire **11**, an inner sheath **12**, a sheath **13**, and a plate body **14** in the first embodiment. FIG. **4** is a cross-sectional view of the bending tube **2***b* near the plate body **14**, taken orthogonal to the longitudinal axis O, in the first embodiment. FIG. **4** is a view of a cross section viewed toward a distal end direction along the longitudinal axis O.

[0055] The wire **11** is configured to be pulled or loosened to bend the bending tube **2***b*.

[0056] In the case where the bending tube **2***b* is bendable in four directions (up, down, left, and right), the bending wire **11** includes an up-bending wire **11***u*, a down-bending wire **11***d*, a left-bending wire **11***l*, and a right-bending wire **11***r*.

[0057] The up-bending wire **11***u* is pulled to thereby bend the bending tube **2***b* in the up direction. The down-bending wire **11***d* is pulled to thereby bend the bending tube **2***b* in the down direction. The left-bending wire **11***l* is pulled to thereby bend the bending tube **2***b* in the left direction. The right-bending wire **11***r* is pulled to thereby bend the bending tube **2***b* in the right direction.

[0058] The bending wires **11***u*, **11***d*, **11***r*, and **11***l* in the respective directions are respectively passed through four inner sheaths **12** (second sheaths) so as to be slidable. Furthermore, the four inner sheaths **12**, through which the bending wires **11***u*, **11***d*, **11***r*, and **11***l* are passed, are respectively passed through four sheaths **13** (first sheaths). Thus, the wires **11** are passed through the sheaths **13** via the inner sheaths **12**.

[0059] The inner sheath **12** may be formed of a thermoplastic elastomer, which is a resin, and may be formed of, for example, PTFE (polytetrafluoroethylene) or PP (polypropylene) based resin, etc.

[0060] The sheath **13** is configured as, for example, a coil sheath having elasticity, and, in a compressed state in the direction of the longitudinal axis O relative to the natural length, is mounted in the endoscope **1**. Note that the sheath **13** configured as a coil sheath has a coil shape over the entire length, but in the drawings, the illustration is simplified as appropriate, and parts other than end portions (such as near the plate body **14**) are shown in a cylindrical shape. However, only a part of the entire length of the sheath **13** may be formed in a coil shape.

[0061] Here, a friction coefficient (first friction coefficient) between the inner sheath **12** and the wire **11** is less than a friction coefficient (second friction coefficient) between the sheath **13** and the wire **11**. Note that by providing the inner sheath **12**, which reduces friction, the use of a lubricant such as silicon oil or carbon powder becomes unnecessary, and the need to ensure watertightness in the insertion portion **2** is eliminated, resulting in a configuration suitable for manufacturing inexpensive endoscopes.

[0062] The plate body **14** having an annular shape is disposed in the proximal end side of the bending tube **2***b*. The plate body **14** is a ring member configured as a separate body from the bending tube **2***b*.

[0063] As shown in FIGS. **2** to **4**, the plate body **14** includes a first annular surface **14***a* facing distally, a second annular surface **14***b* facing proximally, and a hole **14***c* penetrating the first annular surface **14***a* and the second annular surface **14***b*. The second annular surface **14***b* intersects the longitudinal axis O. The second annular surface **14***b* is orthogonal to the longitudinal axis O.

[0064] At least a portion of the sheath **13** is disposed on the proximal end side of the plate body **14**. At least the portion of the sheath **13** is disposed proximally relative to the second annular surface **13***b*. In the examples shown in FIGS. **2** and **3**, the entirety of the sheath **13** is disposed on the proximal end side of the plate body **14**. The sheath **13** is mounted in the endoscope **1**, in a compressed state, as mentioned above. For this reason, a distal end **13***a* of the sheath **13** pushes a periphery part of the hole **14***c*, the periphery part being of the second annular surface **14***b*, by a restoring force produced by the sheath **13** being compressed.

[0065] In other words, the distal end **13***a* of the sheath **13** exerts a normal force against the periphery part of the hole **14***c*, the periphery part being of the second annular surface **14***b*. The distal end **13***a* of the sheath **13** then produces a frictional force in a direction intersecting the longitudinal axis O between the distal end **13***a* of the sheath **13** and the periphery part of the hole

14*c*, the periphery part being of the second annular surface **14***b*.
[0066] At this time, the distal end **13***a* of the sheath **13** stably butts against the second annular surface **14***b*, since the second annular surface **14***b* is orthogonal to the longitudinal axis O. Therefore, the plate body **14** does not require a configuration, such as a long hole, through which the sheath **13** is passed, for causing the distal end **13***a* to stably butt against the second annular surface **14***b*, thus enabling to reduce the thickness of the plate body **14** in the longitudinal axis O direction. Note that, if necessary, the second annular surface **14***b* can be configured to obliquely intersect the longitudinal axis O.
[0067] The plate body **14** is sandwiched between the distal end **13***a* of the sheath **13** and the proximal end side of the bending tube **2***b* (for example, a guide member **16** disposed on the proximal-most end side of a plurality of guide members **16** disposed in the bending tube **2***b*).
[0068] The plate body **14** is neither welded nor bonded to the bending tube **2***b*. The plate body **14** abuts the proximal end side of the bending tube **2***b* (butted against the proximal end side of the bending tube **2***b*), and is fixed to the proximal end side of the bending tube **2***b* by the pressing force received from the sheath **13**. Thus, the plate body **14** functions as a sheath stopper plate that locks the distal end **13***a* of the sheath **13**.
[0069] The wire **11** in a state of being passed through the inner sheath **12** is passed through the hole **14***c*. Thus, a distal end **12***a* of the inner sheath **12** is disposed on the distal end side with respect to the hole **14***c*.
[0070] A part of the inner sheath **12** that is passed through the hole **14***c* of the plate body **14** is bonded to the hole **14***c*. The inner sheath **12** and the plate body **14** are unitized by bonding. The plate body **14** functions as a member to fix the distal end side of the inner sheath **12**. Note that the proximal end side of the inner sheath **12** is disposed and is not fixed in the operation portion **3**.
[0071] Note that here, a configuration is adopted where the distal end side of the inner sheath **12** is bonded and fixed to the hole **14***c* of the plate body **14** disposed in the bending tube **2***b*, and the proximal end side of the inner sheath **12** is left free. In this case, there is an advantage that the manufacturing process for fixing the inner sheath **12** is made easier.
[0072] However, the inner sheath **12** is not limited to this configuration. For example, a configuration can be adopted where the proximal end side of the inner sheath **12** is fixed to a member disposed in the operation portion **3** (such as, for example, a fixing body **31** shown in FIGS. **9** and **10** described later), and the distal end side of the inner sheath **12** is left free. Furthermore, if there is no hindrance in use, the inner sheath **12** can be left free without being fixed.
[0073] The bending tube **2***b* includes a plurality of guide members **16** (see FIG. **8**, etc.) that hold (guide) the wire **11**. The plurality of guide members **16** each serve as a bending piece, for example. The guide member **16** disposed on a proximal-most end side of the bending tube **2***b* is connected to a flexible tube body **2***c***1** of the flexible tube **2***c*, as shown in FIG. **2**.
[0074] The hole **14***c* is provided at a position (first radial position) different from a position (second radial position) where the guide member **16** holds the wire **11**, in a plane orthogonal to the longitudinal axis O.
[0075] The guide member **16** include, for example, a hole **16***a* that functions as a guide hole for the wire **11**. The wire **11** is passed through the hole **16***a* in a movable manner in a direction parallel to the longitudinal axis O. The position of the hole **16***a* in a plane orthogonal to the longitudinal axis O is a position where the guide member **16** holds the wire **11**.
[0076] By differentiating the position of the hole **14***c* in the plate body **14** and the position of the hole **16***a* in the guide member **16** (here, especially positions in a radial direction with the longitudinal axis O at the center) in the plane orthogonal to the longitudinal axis O, a swing angle is produced in the wire **11** entering from the flexible tube **2***c* to the bending tube **2***b*.
[0077] This enables to give a swing angle to the wire **11** without providing an oblique long hole in the plate body **14**. Therefore, the thickness of the plate body **14** in the longitudinal axis O direction can be reduced. This shortens the length of a non-flexible part of the insertion portion **2** in the

longitudinal axis O direction, thereby improving the insertability of the insertion portion **2**.

[0078] In the example shown in FIG. **2**, the hole **14***c* is provided at a position closer to the longitudinal axis O than the hole **16***a* where the guide member **16** holds the wire **11**, in the plane orthogonal to the longitudinal axis O. In this case, the hole **16***a* is located radially outward with respect to the hole **14***c*, with the longitudinal axis O as the center.

[0079] The wire **11** then produces a swing angle to a radially outward side when entering the bending tube **2***b* from the flexible tube **2***c*. It is thus enabled to more reliably avoid contact between the wire **11** and other internal components in the bending tube **2***b*, including when the bending tube **2***b* is bent.

[0080] As shown in FIG. **4**, a width W**14** of a periphery part of the hole **14***c* of the second annular surface **14***b* is greater than or equal to a width W**13** of the distal end **13***a* of the sheath **13**. In this case, the entire surface of the distal end **13***a* of the sheath **13** abuts the second annular surface **14***b*. As a result, the distal end **13***a* of the sheath **13** butts against the second annular surface **14***b* in stable manner.

[0081] FIG. **5** shows a cross-sectional view of a modification of the plate body **14** in the first embodiment, when cut in a plane parallel to the longitudinal axis O.

[0082] The plate body **14** shown in the modification in FIG. **5** includes a groove **14***d* in the periphery part of the hole **14***c* in the second annular surface **14***b* on the proximal end side. The groove **14***d* is configured to be butted against by the distal end **13***a* of the sheath **13**. By providing the groove **14***d*, it is enabled to suppress shifting of the position against where the distal end **13***a* butts, in a direction orthogonal to the longitudinal axis O, thereby stabilizing the butting.

[0083] FIG. **6** is a cross-sectional view showing, from an oblique direction, a configuration of a second plate body **15** disposed in the distal end side of the bending tube **2***b* in the first embodiment.

[0084] The endoscope **1** includes the second plate body **15** disposed in the distal end side of the bending tube **2***b*. The second plate body **15** is disposed to intersect the longitudinal axis O. The second plate body **15** includes a distal-end-side surface **15***a* and a proximal-end-side surface **15***b*. The second plate body **15** is disposed in the distal end side of the bending tube **2***b* such that the distal-end-side surface **15***a* and the proximal-end-side surface **15***b* are orthogonal to the longitudinal axis O, for example.

[0085] Incidentally, in the present embodiment, the up-bending wire **11***u* and the left-bending wire **11***l* are coupled together on the distal end side, and configured of a single wire **11***a*. To put it another way, the single wire **11***a* has a portion that functions as the up-bending wire **11***u* and another portion that functions as the left-bending wire **11***l*.

[0086] Furthermore, the down-bending wire **11***d* and the right-bending wire **11***r* are coupled together on the distal end side, and configured of a single wire **11***b*. To put it another way, the single wire **11***b* has a portion that functions as the down-bending wire **11***d* and another portion that functions as the right-bending wire **11***r*.

[0087] Thus, the bending wire **11** is configured to perform up-down and left-right bending by a total of the two wires **11***a* and **11***b*.

[0088] In such a configuration, the second plate body **15** further includes a plurality of holes penetrating the distal-end-side surface **15***a* and the proximal-end-side surface **15***b*.

[0089] Specifically, the second plate body **15** includes, as the plurality of holes, four holes (but not limited to four, but may be an even number of two, or six, or more) for the wire **11***a* to be passed through. The four holes include a first hole **15***u***1**, a second hole **15***u***2**, a third hole **15***l***2**, and a fourth hole **15***l***1**, in order around the longitudinal axis O (for example, clockwise in FIG. **6**).

[0090] Concerning the wire **11***a*, from the proximal end side of the second plate body **15**, the wire **11***u* is passed, in order, through the first hole **15***u***1**, is further passed through the second hole **15***u***2** from a distal end side of the second plate body **15**, through the third hole **15***l***2** from the proximal end side of the second plate body **15**, and through the fourth hole **15***l***1** from the distal end side of the second plate body **15**, and the wire **11***l* is extended to the proximal end side of the second plate

body **15**.
[0091] FIG. **7** is a cross-sectional view showing the wire **11***a* that is passed through the four holes **15***u***1**, **15***u***2**, **15***l***2**, and **15***l***1** in the second plate body **15** in the first embodiment. FIG. **8** is a cross-sectional view showing the two wires **11***a* and **11***b* that are respectively passed through the two holes **15***u***1** and **15***d***1** in the second plate body **15** in the first embodiment.
[0092] The wire **11***a* has the wire **11***u* passed through the first hole **15***u***1** from the proximal end side of the insertion portion **2**, and is extended as a first part **11***a***1** of the wire **11***a* to the distal-end-side surface **15***a*. The first part **11***a***1** of the wire **11***a* is disposed on the distal-end-side surface **15***a* in a circumferential direction, and is passed through the second hole **15***u***2**.
[0093] The wire **11***a* passed through the second hole **15***u***2** is extended to the proximal-end-side surface **15***b* as a second part **11***a***2**. The second part **11***a***2** is disposed on the proximal-end-side surface **15***b* in the circumferential direction, and is passed through the third hole **15***l***2**.
[0094] The wire **11***a* passed through the third hole **15***l***2** is extended to the distal-end-side surface **15***a* as a third part **11***a***3**. The third part **11***a***3** is disposed on the distal-end-side surface **15***a* in the circumferential direction, and is passed through the fourth hole **15***l***1**.
[0095] Concerning the wire **11***a* passed through the fourth hole **15***l***1**, the wire **11***l* is extended to the proximal end side of the insertion portion **2**.
[0096] Also, the second plate body **15** includes, as the plurality of holes, four holes (but not limited to four, but may be an even number of two, or six, or more) for the wire **11***b* to be passed through. The four holes include a first hole **15***d***1**, a second hole **15***d***2**, a third hole **15***r***2**, and a fourth hole **15***r***1**, in order around the longitudinal axis O (for example, clockwise in FIG. **6**).
[0097] Concerning the wire **11***b*, from the proximal end side of the second plate body **15**, the wire **11***d* is passed through the first hole **15***d***1**, is further passed through the second hole **15***d***2**, the third hole **15***r***2**, and the fourth hole **15***r***1** in order, and the wire **11***r* is extended to the proximal end side of the second plate body **15**.
[0098] Similar to the wire **11***a* shown in FIG. **7**, and although illustration of the cross-sectional view is omitted, the wire **11***b* has the wire **11***d* passed through the first hole **15***d***1** from the proximal end side of the insertion portion **2**, and is extended as a first part **11***b***1** of the wire **11***b* (see FIG. **6**) to the distal-end-side surface **15***a*. The first part **11***b***1** of the wire **11***b* is disposed on the distal-end-side surface **15***a* in the circumferential direction, and is passed through the second hole **15***d***2**.
[0099] The wire **11***b* passed through the second hole **15***d***2** is extended to the proximal-end-side surface **15***b* as a second part **11***b***2** (see FIG. **6**). The second part **11***b***2** is disposed on the proximal-end-side surface **15***b* in the circumferential direction, and is passed through the third hole **15***r***2**.
[0100] The wire **11***b* passed through the third hole **15***r***2** is extended to the distal-end-side surface **15***a* as a third part **11***b***3** (see FIG. **6**). The third part **11***b***3** is disposed on the distal-end-side surface **15***a* in the circumferential direction, and is passed through the fourth hole **15***r***1**.
[0101] Concerning the wire **11***b* passed through the fourth hole **15***r***1**, the wire **11***r* is extended to the proximal end side of the insertion portion **2**.
[0102] Thus, the wires **11***a* and **11***b* are each disposed in the second plate body **15** in an M-shape. Note that the wires **11***a* and **11***b* formed in the second plate body **15** in an M-shape may be prepared in advance, and the bending wires **11** may be unitized. In this case, the unitized wires **11** are assembled into the endoscope **1**.
[0103] As shown in FIG. **8**, the proximal-end-side surface **15***b* of the second plate body **15** is pressed against the guide member **16**, which also serves as a bending piece, that is disposed on a distal-most end side. By bringing the wires **11***a* and **11***b* into a state of being applied with tension in a tensile direction, the second plate body **15** is fixed to the guide member **16**. In this case, neither a work of bonding nor a work of welding the second plate body **15** to the guide member **16** is required, thus enabling to reduce manufacturing costs.
[0104] In addition, since the wires **11***a* and **11***b* are in the state of being applied with tension, a frictional force acts between the wire **11***a* and the second plate body **15** and a frictional force acts

between the wire **11***b* and the second plate body **15** at the M-shaped part.
[0105] For this reason, pulling the wire **11***u* does not result in the wire **11***l* being drawn toward the distal end side, and pulling the wire **11***l* does not result in the wire **11***u* being drawn toward the distal end side. Similarly, pulling the wire **11***d* does not result in the wire **11***r* being drawn toward the distal end side, and pulling the wire **11***r* does not result in the wire **11***d* being drawn toward the distal end side. Thus, the second plate body **15** functions as a wire stop plate.
[0106] With the configuration where the single wire **11***a* serves to function as the up-bending wire **11***u* and the left-bending wire **11***l*, and the single wire **11***b* serves to function as the down-bending wire **11***d* and the right-bending wire **11***r*, the need for the work to fix each of distal ends of the four wires to the distal end side of the bending tube **2***b* (or the proximal end side of the distal end portion **2***a*) is eliminated, thereby enabling to reduce manufacturing costs. Furthermore, by unitizing the two wires **11***a* and **11***b* and the second plate body **15**, it is enabled to shorten assembly time and reduce assembly costs.
[0107] Note that, although the configuration was described here that uses the two wires **11***a* and **11***b* to perform four-directional bending, a general configuration of using individually provided four bending wires to perform four-directional bending can also be adopted for the endoscope **1**.
[0108] FIG. **9** is a perspective view showing the fixing body **31** disposed in the operation portion **3** in the first embodiment. FIG. **10** is a cross-sectional view showing a portion of a structure of the fixing body **31** in the first embodiment.
[0109] The operation portion **3** includes an operation portion body **3***a*, which also serves as an exterior. The fixing body **31** is fixed to the operation portion body **3***a* using a screw **32** or the like. The fixing body **31** bundles internal components, such as the treatment instrument channel **21**, the gas/liquid feeding channel **22**, and a signal cable connected to the image pickup device, and fixes positions of these components in the operation portion **3**.
[0110] The fixing body **31** includes a receiving surface **31***a*. The receiving surface **31***a* is configured, for example, as a surface orthogonal to the longitudinal axis O. At least a portion of the sheath **13** is disposed on the distal end side of the receiving surface **31***a*.
[0111] In the example shown in FIGS. **9** and **10**, the entirety of the sheath **13** is disposed on the distal end side of the receiving surface **31***a*. In this case, a proximal end **13***b* of the sheath **13** is pressed against the receiving surface **31***a*. The inner sheath **12** is extended from the proximal end **13***b* of the sheath **13** to the proximal end side, and a proximal end **12***b* (see FIG. **9**) of the inner sheath **12** is disposed on the proximal end side with respect to the fixing body **31**. The proximal end side of the inner sheath **12** is not fixed in the operation portion **3**, as mentioned above.
[0112] The sheath **13** merely has the one end pressed against the second annular surface **14***b* of the plate body **14** and the other end pressed against the receiving surface **31***a*, and is not fixed to any other member.
[0113] The sheath **13** is mounted in the endoscope **1**, in a compressed state, as mentioned above. For this reason, the proximal end **13***b* of the sheath **13** pushes the receiving surface **31***a* of the fixing body **31** by the restoring force of the sheath **13** produced by being compressed.
[0114] In other words, the proximal end **13***b* of the sheath **13** exerts a normal force against the receiving surface **31***a*. The proximal end **13***b* of the sheath **13** then produces a frictional force with the receiving surface **31***a* in a direction intersecting the longitudinal axis O.
[0115] At this time, since the receiving surface **31***a* is orthogonal to the longitudinal axis O, the proximal end **13***b* of the sheath **13** stably butts against the receiving surface **31***a*. Note that, if necessary, the receiving surface **31***a* can be configured to obliquely intersect the longitudinal axis O.
[0116] The fixing body **31** further includes a pressing structure portion **31***b*. The pressing structure portion **31***b* is provided on the distal end side of the receiving surface **31***a* of the fixing body **31**. The pressing structure portion **31***b* abuts a side surface of the sheath **13** to prevent buckling of the sheath **13**. This prevents buckling of the sheath **13**, as well as the inner sheath **12** and the wire **11**

that are passed through the sheath **13**, thus causing no hindrance to the bending operation with the bending operation knob **6**.

[0117] A filling rate of the internal components, such as the treatment instrument channel **21**, in the inside of the flexible tube **2***c* is higher than a filling rate in the inside of the operation portion **3**. The other internal components then prevent the sheath **13** from shifting in the direction orthogonal to the longitudinal axis O in the flexible tube **2***c*. For this reason, no structure portion corresponding to the pressing structure portion **31***b* is provided in the flexible tube **2***c*. In contrast, the operation portion **3**, which has a filling rate of internal components lower than that of the flexible tube **2***c*, has more space to spare, and therefore is provided with the pressing structure portion **31***b*.

[0118] However, the pressing structure portion **31***b* of the operation portion **3** can be omitted if the sheath **13**, the inner sheath **12**, and the wire **11** do not shift in the direction orthogonal to the longitudinal axis O.

[0119] FIG. **11** is a view showing how a position to cut the sheath **13** is set according to an actual length of the flexible tube body **2***c***1** in the first embodiment.

[0120] The actual length of the flexible tube body **2***c***1** that the flexible tube **2***c* includes may differ from a design length due to product variation.

[0121] The sheath **13** is manufactured by cutting a long coil sheath material. If the actual length of the flexible tube **2***c* differs from the design length, cutting the sheath **13** at the design length may not result in the sheath **13** in a properly compressed state when mounted in the endoscope **1**.

[0122] Accordingly, the sheath **13** is cut and formed as shown in FIG. **11** to match the actual length of the flexible tube body **2***c***1**.

[0123] In FIG. **11**, one end of the flexible tube body **2***c***1** manufactured is aligned with a start line LS, and a distal end of the coil sheath material is aligned with the start line LS.

[0124] The design length of the flexible tube body **2***c***1** is assumed to be a length from the start line LS to a flexible-tube design line LD**1**. However, the actual length of the flexible tube body **2***c***1** in the illustrated example is larger than the length from the start line LS to the flexible-tube design line LD**1** by a length L**1**.

[0125] On the other hand, the design length of the sheath **13** is assumed to be a length from the start line LS to a sheath design line LD**2**. In this case, corresponding to the fact that the flexible tube body **2***c***1** is longer than the design length by the length L**1**, a cutting line LC to cut the sheath **13** is set at a position longer than the sheath design line LD**2** by the length L**1**, to cut the long coil sheath material.

[0126] By adopting such a manufacturing method, it is enabled that the sheath **13** maintains a properly compressed state when mounted in the endoscope **1**, even when there is product variation in the length of the flexible tube body **2***c***1**.

[0127] FIG. **12** is a side view of the operation mechanism **33** disposed in the operation portion **3** in the first embodiment. FIG. **13** is a partial perspective view showing the operation mechanism **33** of the first embodiment from a direction of an arrow A**1** in FIG. **12**.

[0128] The operation portion **3** includes the operation mechanism **33** configured to receive an operation to pull or loosen the wire **11**. The proximal end side of the wire **11** is fixed to the operation mechanism **33**.

[0129] The operation mechanism **33** includes a pulley **34** and a pulley **35**. The pulley **34** pivots around a rotation axis R in conjunction with a pivoting operation of the up-down bending operation knob **6***a*. The pulley **35** pivots around the rotation axis R in conjunction with a pivoting operation of the left-right bending operation knob **6***b*.

[0130] Note that, although an example has been shown here where the operation mechanism **33** is in conjunction with the bending operation knob **6**, the operation mechanism **33** is not limited to this configuration. For example, the operation mechanism **33** can be configured to be electrically pivoted by an actuator or the like in response to a joystick operation, or may adopt any other appropriate configuration.

[0131] Around the pulley **34**, the up-bending wire **11***u* and the down-bending wire **11***d* are wound, and to the pulley **34**, the proximal end side of the wire **11***u* and the proximal end side of the wire **11***d* are fixed. Around the pulley **35**, the right-bending wire **11***r* and the left-bending wire **11***l* are wound, and to the pulley **35**, the proximal end side of the wire **11***r* and the proximal end side of the wire **11***l* are fixed.
[0132] The pulley **34** includes an up-bending wire winding chamber **34***u* and a down-bending wire winding chamber **34***d*, which are located at different positions in the direction of the rotation axis R. The up-bending wire winding chamber **34***u* and the down-bending wire winding chamber **34***d* are separated from each other by, for example, a flange or the like interposed therebetween.
[0133] The pulley **35** includes a right-bending wire winding chamber **35***r* and a left-bending wire winding chamber **35***l*, which are located at different positions in the direction of the rotation axis R. The right-bending wire winding chamber **35***r* and the left-bending wire winding chamber **35***l* are separated from each other by, for example, a flange or the like interposed therebetween.
[0134] For example, the wire **11***r* and the wire **11***l* are crossed on the proximal end side of pulley **35**, as shown in FIG. **13**. The wire **11***r* is crossed, is then wound once around the right-bending wire winding chamber **35***r*, and is fixed to a fixation portion **35***a* of the pulley **35** by a first knock pin **36**. The wire **11***l* is crossed, is then wound once around the left-bending wire winding chamber **35***l*, and is fixed to the fixation portion **35***a* of the pulley **35** by a second knock pin **36**. With such a configuration, the wires **11***r* and **11***l* do not overlap, and each can transmit a pulling force and each can loosen.
[0135] The wires **11***u* and **11***d* are fixed to the pulley **34** in a similar manner as the wires **11***r* and **11***l*. That is, the wires **11***u* and **11***d* are crossed on the proximal end side of the pulley **34**. The wire **11***u* is crossed, is the wound once around the up-bending wire winding chamber **34***u*, and is fixed to a fixation portion **34***a* of the pulley **34** by a third knock pin **36**. The wire **11***d* is crossed, is then wound once around the down-bending wire winding chamber **34***d*, and is fixed to the fixation portion **34***a* of the pulley **34** by a fourth knock pin **36**. With such a configuration, the wires **11***u* and **11***d* do not overlap, and each can transmit a pulling force and each can loosen.
[0136] Here, standard general-purpose parts may be used as the first to fourth knock pins **36**. By using general-purpose parts, the manufacturing cost of endoscope **1** can be reduced compared to a case of using dedicated parts.
[0137] As described with reference to FIGS. **6** to **8**, when the wire **11***u* and the wire **11***l* are configured of the single wire **11***a*, the wire **11***a* has the one end (the proximal end of the wire **11***u*) fixed to the pulley **34**, is extended to the distal end side, is passed through the flexible tube **2***c* and the bending tube **2***b*, is disposed in the second plate body **15** in an M-shape to be extended to the proximal end side, is passed through the bending tube **2***b* and the flexible tube **2***c*, and has the other end (the proximal end of the wire **11***l*) fixed to the pulley **35**.
[0138] Similarly, when the wire **11***d* and the wire **11***r* are configured of the single wire **11***b*, the wire **11***b* has the one end (the proximal end of the wire **11***d*) fixed to the pulley **34**, is extended to the distal end side, is passed through the flexible tube **2***c* and the bending tube **2***b*, is disposed in the second plate body **15** in an M-shape to be extended to the proximal end side, is passed through the bending tube **2***b* and the flexible tube **2***c*, and has the other end (the proximal end of the wire **11***r*) fixed to the pulley **35**.
[0139] FIG. **14** is a view for illustrating a process of fixing the bending wire **11** to the operation mechanism **33** in the first embodiment.
[0140] As mentioned above, the wire **11** is fixed in the endoscope **1** in a state of being applied with tension. The fixing of the wire **11** inside the endoscope **1** is performed, for example, by the following process.
[0141] A bending tube base **37**, the fixing body **31**, and the operation mechanism **33** that are provided on the distal end side of the operation portion **3** are each fixed to a jig at positions separated by respective specified distances from each other.

[0142] Furthermore, the fixation portions **34***a* and **35***a* of the operation mechanism **33** are pressed against a rotation-stopping jig **38** to hold the pulleys **34** and **35** in a non-rotating state.
[0143] The proximal end side of the wire **11** is then wound around a winding jig **39**, and the winding jig **39** is rotated to wind the proximal end side of the wire **11**.
[0144] When the wire **11** is wound by the jig **39**, the bending tube **2***b* bends in accordance with the winding amount. Once the bending tube **2***b* is in a maximum bending state, the winding force of the jig **39** is adjusted to obtain a state where the wire **11** is tensioned at a constant tension.
[0145] Once this state is obtained, the knock pin **36** is press-fitted into each of the fixation portions **34***a* and **35***a* to fix the proximal end side of the wire **11**.
[0146] By doing such a work for each of the wires **11***u*, **11***d*, **11***r*, and **11***l*, a state can be obtained in which a constant tension is applied to each of the wires **11***u*, **11***d*, **11***r*, and **11***l* even when the bending tube **2***b* is in a straight-line state.
[0147] Here, if hypothetically the wire **11** is not applied with a constant tension, it exhibits slack. If this is the case, even if the bending operation starts, the bending will not start until the wire **11** is taut, thus reducing the responsiveness of the bending to the user operation.
[0148] In contrast, the wire **11** fixed in the above-mentioned process is in a state of being applied with a constant tension, thus eliminating slack and enabling bending to be performed with high responsiveness to user operation.
[0149] FIG. **15** is a view showing that the sheath **13** is held in a state of pressing the plate body **14** and the fixing body **31** in the first embodiment.
[0150] The sheath **13** has the distal end **13***a* pressed against the second annular surface **14***b* of the plate body **14**, and the proximal end **13***b* pressed against the receiving surface **31***a* of the fixing body **31**, thereby pushing the plate body **14** and the fixing body **31**, as shown by both arrows.
[0151] As a configuration that operates in this manner, an example was mentioned above in which the sheath **13** is a coil sheath in a compressed state. However, the sheath **13** is not limited to a coil sheath and can have any other configuration having elasticity. For example, the sheath **13** can be a tubular member formed of elastic rubber.
[0152] Furthermore, the sheath **13** does not have to be configured of a common elastic material (a material having compressibility). For example, a small-diameter endoscope may adopt a thin-walled metal pipe as a tubular member that configures a channel. Even a metal pipe, if capable of being mounted in the endoscope **1** in a state where both ends of the metal pipe are pressed, will induce a normal force against the plate body **14** and the fixing body **31**, and produce a frictional force in a direction intersecting the longitudinal axis O between the plate body **14** and the fixing body **31**. Therefore, the sheath **13** may be a cylindrical member, such as a metal pipe, formed of a rigid material.
[0153] FIG. **16** is a cross-sectional view showing a relationship between the sheath **13** and the wire **11** when the inner sheath **12** is not provided in the first embodiment.
[0154] The wire **11** moves in the direction of the longitudinal axis O when pulled or loosened. On the other hand, the sheath **13** is sandwiched between the plate body **14** and the fixing body **31**, as shown in FIG. **15**, and does not move in the direction of the longitudinal axis O. Therefore, when the wire **11** is pulled or loosened, a relative position change in the direction of the longitudinal axis O will occur between the wire **11** and the sheath **13**.
[0155] If hypothetically the inner sheath **12** is not provided, as shown in FIG. **16**, when the positions of the wire **11** and the sheath **13** change relative to each other, the wire **11** contacts and causes friction on the surface of the core wire of the sheath **13**, which is configured as a coil sheath, and the wire **11** may wear. Furthermore, when the positions of the wire **11** and the sheath **13** change relative to each other, the wire **11** contacts and causes friction on an edge of an end portion of the sheath **13**, and the wire **11** may wear.
[0156] In contrast, in the present embodiment, as mentioned above, the inner sheath **12** has the distal end **12***a* disposed on the distal end side with respect to the hole **14***c* of the plate body **14**, and

has the proximal end **12***b* disposed on the proximal end side with respect to the proximal end **13***b* of the sheath **13**.
[0157] In other words, the inner sheath **12** is passed through and over the entire length of the sheath **13**, ensuring that the wire **11** does not come into direct contact with the sheath **13**. Furthermore, as mentioned above, the friction coefficient between the inner sheath **12** and the wire **11** is less than the friction coefficient between the sheath **13** and the wire **11**.
[0158] This prevents the wire **11** from contacting the surface or edge of the core wire of the sheath **13** and wearing. Furthermore, an increase in the amount of force due to the friction can be suppressed when the wire **11** is pulled.
[0159] In addition, the distal end **12***a* of the inner sheath **12** is disposed on the distal end side with respect to the hole **14***c*. The wire **11** then not only does not contact the sheath **13**, but also does not contact the plate body **14**. This enables to prevent the wire **11** from contacting the surface or edge of the plate body **14** and wearing.
[0160] The proximal end **12***b* of the inner sheath **12** is disposed on the proximal end side with respect to the fixing body **31**, as mentioned above. The wire **11** then does not contact the fixing body **31** either. This enables to prevent the wire **11** from contacting the surface or edge of the fixing body **31** and wearing.
[0161] According to the first embodiment, the sheath **13** is not fixed, only having the one end pushing the plate body **14** and the other end pushing the fixing body **31**. Therefore, there is no need for skilled workers to use a jig to conduct the welding work and bonding work when mounting the sheath **13** in the endoscope **1**. For this reason, the endoscope **1**, which includes the sheath **13** through which the bending wire **11** is passed, is suitable for inexpensive manufacturing.
[0162] Furthermore, since the plate body **14** is fixed by the pressing force of the sheath **13**, and the second plate body **15** is fixed by the tension of the wire **11**, welding and bonding are unnecessary for both the plate body **14** and the second plate body **15**. Thus, it is enabled to manufacture the endoscope **1** more inexpensively.
[0163] In addition, by differentiating the position of the hole **14***c* in the plate body **14** and the position of the hole **16***a* in the guide member **16**, the wire **11** is given a swing angle. For this reason, it is enabled to dispose the wire **11** in different manners in the flexible tube **2***c* and the bending tube **2***b*, thereby increasing the degree of freedom in the disposition of the internal components.
SECOND EMBODIMENT
[0164] FIG. **17** is a view showing a configuration example of the sheath **13** in a second embodiment of the present disclosure. In the second embodiment, parts similar to those in the first embodiment are attached with the same symbols, and descriptions thereof are omitted as appropriate. In the second embodiment, points different from the first embodiment are mainly described.
[0165] The sheath **13** has a portion **13***x* disposed on the proximal end side of the second annular surface **14***b* of the plate body **14**, and has another portion **13***y* disposed on the distal end side of the second annular surface **14***b* of the plate body **14**. The portion **13***x* is longer than the other portion **13***y* in the direction of the longitudinal axis O.
[0166] A flange **13***f* protruding radially outward is provided at a distal end of the portion **13***x* of the sheath **13**. Therefore, the position where the flange **13***f* is provided is in the distal end side of the sheath **13**, but not at the distal end **13***a* of the entire sheath **13**.
[0167] The flange **13***f* of the sheath **13** abuts the second annular surface **14***b* to push the plate body **14**. The surface of the flange **13***f*, which abuts the second annular surface **14***b*, is a distal end **13***a'* of the portion **13***x* of the sheath **13**. Therefore, the distal end **13***a'* of the portion **13***x* of the sheath **13** is the section that pushes the plate body **14**.
[0168] Note that a configuration can be employed to provide, instead of the flange **13***f*, a projection or the like protruding radially outward from the sheath **13**, to push the plate body **14** by the

projection or the like.

[0169] If the sheath **13** is formed of a non-compressible material, a configuration may be employed to provide a separate elastic member, so that the elastic member presses the flange **13***f* against the plate body **14**.

[0170] Furthermore, instead of the configuration to use the flange **13***f* to push the plate body **14**, a configuration is also employable to bond a portion of an outer circumferential surface of the sheath **13** integrally with the plate body **14**, so that the sheath **13** presses the plate body **14**.

[0171] According to the second embodiment, an effect almost the same as that of the above-mentioned first embodiment is yielded, even when a part that is not the distal end **13***a* of the entire sheath **13**, i.e., the distal end **13***a'* of the portion **13***x* of the sheath **13**, which is disposed on the proximal end side of the plate body **14**, is configured to push the plate body **14**.

THIRD EMBODIMENT

[0172] FIGS. **18** and **19** show a third embodiment of the present disclosure. In the third embodiment, parts similar to those in the first and second embodiments are attached with the same symbols, and descriptions a thereof are omitted as appropriate. In the third embodiment, points different from the first and second embodiments are mainly described.

[0173] As mentioned above, by differentiating the position of the hole **14***c* in the plate body **14** and the position of the hole **16***a* in the guide member **16**, the wire **11** can be given a swing angle.

[0174] FIG. **18** is a view showing a first example of the swing angle of the wire **11** in the third embodiment.

[0175] In the plane orthogonal to the longitudinal axis O, the hole **14***c* in the plate body **14** is provided at a position closer to the longitudinal axis O than the position where the guide member **16** holds the wire **11**, i.e., than the hole **16***a* in the guide member **16**. The swing angle of the first example shown in FIG. **18** is almost the same as the swing angle of the wire **11** described in the first embodiment.

[0176] In this case, the swing angle of the wire **11** is a swing angle to a radially outward side as the wire **11** extends from the flexible tube **2***c* side toward the bending tube **2***b* side. By swinging the wire **11** to a radially outward side to widen the internal space of the bending tube **2***b*, it is enabled to increase the degree of freedom in the disposition of the internal components in the bending tube **2***b*.

[0177] FIG. **19** is a view showing a second example of the swing angle of the wire **11** in the third embodiment.

[0178] In the plane orthogonal to the longitudinal axis O, the hole **14***c* in the plate body **14** is provided at a position farther from the longitudinal axis O than the position where the guide member **16** holds the wire **11**, i.e., than the hole **16***a* in the guide member **16**.

[0179] In this case, the swing angle of the wire **11** is a swing angle to a radially inward side as the wire **11** extends from the flexible tube **2***c* side to the bending tube **2***b* side. The bending tube **2***b* tends to have a thicker outer skin and therefore a larger outer diameter. For this reason, by swinging the wire **11** to a radially inward side to narrow the inside of the bending tube **2***b*, it is enabled to suppress the outer diameter of the bending tube **2***b* from becoming larger.

[0180] Note that the above showed an example of shifting the positions of the hole **14***c* and the hole **16***a* in the radial direction from each other, but no limitation is placed thereon and positions of the hole **14***c* and the hole **16***a* in the circumferential direction can be shifted from each other as well. In this case, it is facilitated to adjust the disposition of the internal components.

[0181] According to the third embodiment, an effect almost the same as that of the above-mentioned first and second embodiments is yielded.

[0182] According to the third embodiment, by providing a swing angle to the wire **11**, it is enabled to increase the degree of freedom in the disposition of the internal components, facilitate the adjustment of the disposition of the internal components, or suppress the outer diameter of the bending tube **2***b* from becoming larger.

FOURTH EMBODIMENT

[0183] FIG. **20** shows a fourth embodiment of the present disclosure. In the fourth embodiment, parts similar to those in the first to third embodiments are attached with the same symbols, and descriptions thereof are omitted as appropriate. In the fourth embodiment, points different from the first to third embodiments are mainly described.

[0184] FIG. **20** is a view showing a configuration example of the sheath **13** in a fourth embodiment.

[0185] The sheath **13** has a portion **13***x*′ disposed on the distal end side of the receiving surface **31***a* of the fixing body **31**, and has another portion **13***y*′ disposed on the proximal end side of the receiving surface **31***a* of the fixing body **31**. The portion **13***x*′ is longer than the other portion **13***y*′ in the direction of the longitudinal axis O.

[0186] A flange **13***g* protruding radially outward is provided at a proximal end of the portion **13***x*′ of the sheath **13**. Therefore, the position where the flange **13***g* is provided is on the proximal end side of the sheath **13**, but not at the proximal end **13***b* of the entire sheath **13**.

[0187] The flange **13***g* of the sheath **13** abuts the receiving surface **31***a* of the fixing body **31** to push the fixing body **31**. The surface of the flange **13***g*, which abuts the receiving surface **31***a*, is the proximal end **13***b*′ of the portion **13***x*′ of the sheath **13**. Therefore, the proximal end **13***b*′ of the portion **13***x*′ of the sheath **13** is the section that pushes the fixing body **31**.

[0188] Note that a configuration can be employed to provide, instead of the flange **13***g*, a projection or the like protruding radially outward from the sheath **13**, to push the fixing body **31** by the projection or the like.

[0189] If the sheath **13** is formed of a non-compressible material, a configuration may be employed to provide a separate elastic member, so that the elastic member presses the flange **13***g* against the fixing body **31**.

[0190] Furthermore, the sheath **13** can be configured to include both the flange **13***f* shown in FIG. **17** and the flange **13***g* shown in FIG. **20**.

[0191] According to the fourth embodiment, an effect almost the same as that of the above-mentioned first to third embodiments is yielded, even when a part that is not the proximal end **13***b* of the entire sheath **13**, i.e., the proximal end **13***b*′ of the portion **13***x* of the sheath **13**, which is disposed on the distal end side of the fixing body **31**, is configured to push the fixing body **31**.

FIFTH EMBODIMENT

[0192] FIG. **21** shows a fifth embodiment of the present disclosure. In the fifth embodiment, parts similar to those in the first to fourth embodiments are attached with the same symbols, and descriptions thereof are omitted as appropriate. In the fifth embodiment, points different from the first to fourth embodiments are mainly described.

[0193] FIG. **21** is a diagram showing a configuration example of a part of the second annular surface **14***b* of the plate body **14**, where the distal end **13***a* of the sheath **13** abuts, in a fifth embodiment.

[0194] As mentioned above, the plate body **14** is formed as a ring member for the internal components to be passed through. In this case, a ring width of the plate body **14** in the plane orthogonal to the longitudinal axis O should be small so as not to interfere with the passing through of the internal components.

[0195] However, the plate body **14** receives a force pushed from the distal end **13***a* of the sheath **13**. For this reason, the width W**14** of the part of the plate body **14**, where the distal end **13***a* abuts, can be greater than or equal to the width (outer diameter) W**13** of the distal end **13***a* of the sheath **13**. In the example shown in column A of

[0196] FIG. **21**, the plate body **14** is configured such that W**14**≥W**13** (in the illustrated exemplary example, W**14**>W**13**). However, the width of the plate body **14** other than the part where the distal end **13***a* abuts is less than W**14**. Therefore, the part of the plate body **14**, which has the width W**14**, is shaped to protrude radially inward.

[0197] According to the configuration example shown in column A of FIG. **21**, both improved passability of the internal components and reception of a stable pressing force by the plate body **14** from the distal end **13***a* of the sheath **13** can be achieved.
[0198] An example shown in column B of FIG. **21** is one in which the width of the part of the plate body **14**, where the distal end **13***a* abuts, is smaller than that in column A of FIG. **21**. In this case, the entirety of the distal end **13***a* does not abut the second annular surface **14***b* of the plate body **14**, but only a portion of the distal end **13***a* abuts the second annular surface **14***b*.
[0199] In the configuration shown in column A of FIG. **21**, the distal end **13***a* is pressed against the second annular surface **14***b* in a circular shape, while in the configuration shown in column B of FIG. **21**, the distal end **13***a* is pressed against the second annular surface **14***b* in a D shape (shape of a portion of a circular arc). If the plate body **14** can receive a stable pressing force from the distal end **13***a* of the sheath **13**, the configuration shown in column B of FIG. **21** can be adopted.
[0200] According to the configuration example shown in column B of FIG. **21**, the filling rate of the internal components that are passed through the plate body **14** can be decreased. Furthermore, the materials used to manufacture the plate body **14** can be reduced.
[0201] On the other hand, the example shown in column C of FIG. **21** is an example in which the width of the plate body **14** in the plane orthogonal to the longitudinal axis O is a constant width greater than or equal to the width of the distal end **13***a* of the sheath **13**. If there is enough space in the flexible tube **2***c* with no hindrance for passage through of the internal components, the configuration shown in column C of FIG. **21** can be adopted. According to the configuration example shown in column C of FIG. **21**, the aligning work to the part having a wide ring width is unnecessary when providing four holes **14***c* in the plate body **14**.
[0202] According to the fifth embodiment, an effect almost the same as that of the above-mentioned first to four embodiments is yielded.
[0203] According to the fifth embodiment, it is enabled to select the plate body **14** with an appropriate configuration according to the stability of pressing the sheath **13** against the plate body **14**, the need to decrease the filling rate of the internal components, the ease of the work to provide the hole **14***c* in the plate body **14**, etc.
[0204] Note that the present disclosure is not limited to the above-mentioned embodiments as-is. The present disclosure can be embodied by modifying the components to an extent not departing from the gist of the present disclosure, at the implementation stage. Also, a plurality of components disclosed in the above-mentioned embodiments can be combined as appropriate to form various aspects of the present disclosure. For example, some components may be deleted from all the components disclosed in the embodiments. Furthermore, components of different embodiments may be combined as appropriate. Thus, it is of course possible to make various variations and applications within the scope not departing from the essence of the present disclosure.

Claims

1. An endoscope comprising: a bending tube extending along a longitudinal axis from a proximal end side to a distal end side; a wire configured to bend the bending tube; a first sheath through which the wire is passed; and a plate body having an annular shape and disposed in the proximal end side of the bending tube, wherein the plate body comprising: a first annular surface facing distally; a second annular surface facing proximally and intersecting the longitudinal axis; and a hole penetrating the first annular surface and the second annular surface, and through which the wire is passed, and at least a portion of the first sheath is disposed proximally relative to the second annular surface, and a distal end of at least the portion of the first sheath is configured to push a periphery of the hole on the second annular surface.
2. The endoscope according to claim 1, wherein the first sheath has elasticity, and the distal end of at least the portion of the first sheath is configured to push the periphery of the hole on the second

annular surface, by a restoring force produced by the first sheath compressed in a direction of the longitudinal axis.

3. The endoscope according to claim 2, wherein the first sheath is a coil sheath.

4. The endoscope according to claim 1, further comprising: a second sheath passed through the first sheath, and through which the wire is passed, wherein the second sheath is passed through the hole, and a distal end of the second sheath extends distally relative to the hole.

5. The endoscope according to claim 4, wherein a portion of the second sheath is bonded to the hole.

6. The endoscope according to claim 4, wherein a first friction coefficient between the second sheath and the wire is less than a second friction coefficient between the first sheath and the wire.

7. The endoscope according to claim 6, wherein the second sheath is formed of resin, and the second annular surface extends orthogonal to the longitudinal axis.

8. The endoscope according to claim 1, wherein an entirety of the first sheath is disposed proximally relative to the plate body, and a distal end of the first sheath pushes the periphery of the hole on the second annular surface.

9. The endoscope according to claim 8, wherein a width of the periphery of the hole on the second annular surface, is greater than or equal to a width of the distal end of the first sheath, and an entire surface of the distal end of the first sheath abuts the second annular surface.

10. The endoscope according to claim 1, wherein the bending tube includes a guide member that holds the wire, and the hole is provided at a first radial position different from a second radial position where the guide member holds the wire.

11. The endoscope according to claim 10, wherein the first radial position is closer to the longitudinal axis than the second radial position.

12. The endoscope according to claim 10, wherein the first radial position is farther from the longitudinal axis than the second radial position.

13. The endoscope according to claim 1, further comprising: a flexible tube provided proximally relative to the bending tube; and an operation portion provided proximally relative to the flexible tube, wherein the operation portion includes a fixing body including a receiving surface orthogonal to the longitudinal axis, and at least the portion of the first sheath is disposed distally relative to the receiving surface, and a proximal end of at least the portion is configured to push the receiving surface.

14. The endoscope according to claim 13, wherein an entirety of the first sheath is disposed distally relative to the receiving surface, and a proximal end of the first sheath is configured to push the receiving surface.

15. The endoscope according to claim 13, wherein the fixing body includes a pressing structure arranged distally relative to the receiving surface, the pressing structure configured to abut a side surface of the first sheath to prevent buckling of the first sheath.

16. The endoscope according to claim 13, wherein the operation portion includes an operation mechanism configured to receive an operation to pull or loosen the wire, and a proximal end side of the wire is fixed to the operation mechanism.

17. The endoscope according to claim 16, wherein the operation mechanism includes a pulley around which the wire is wound, and the proximal end side of the wire is fixed to the pulley by a knock pin.

18. The endoscope according to claim 1, further comprising: a second plate body disposed on the distal end side of the bending tube, wherein the second plate body includes a first hole, a second hole, a third hole, and a fourth hole in order around the longitudinal axis, the first to fourth holes penetrating a distal-end-side surface and a proximal-end-side surface of the second plate body, and the wire is passed, in order, through the first hole from a proximal end side of the second plate body, through the second hole from a distal end side of the second plate body, through the third hole from the proximal end side of the second plate body, and through the fourth hole from the distal end side of the second plate body.

19. An endoscope comprising: a bending tube extending along a longitudinal axis from a proximal end side to a distal end side; a wire configured to bend the bending tube; a first sheath through which the wire is passed; and a plate body having an annular shape and disposed in the proximal end side of the bending tube, wherein the plate body comprising: a first annular surface facing distally; a second annular surface facing proximally and intersecting the longitudinal axis; and a hole penetrating the first annular surface and the second annular surface, and through which the wire is passed, and at least a portion of the first sheath is disposed proximally relative to second annular surface, and a distal end of at least the portion of the first sheath is configured to produce a frictional force in a direction intersecting the longitudinal axis between the distal end of at least the portion and a periphery of the hole on the second annular surface.

20. An endoscope comprising: a bending tube extending along a longitudinal axis from a proximal end side to a distal end side; a wire configured to bend the bending tube; a first sheath; a second sheath passed through the first sheath, and through which the wire is passed; and a plate body having an annular shape and disposed in the proximal end side of the bending tube, wherein the plate body includes: a first annular surface facing distally; a second annular surface facing proximally; and a hole penetrating the first annular surface and the second annular surface, and through which the wire is passed, at least a portion of the first sheath is disposed proximally relative to the second annular surface, and a distal end of at least portion of the first sheath is configured to push a periphery of the hole on the second annular surface, and the wire and the second sheath are passed through the hole, and a distal end of the second sheath is disposed distally relative to the hole.